



ST BENEDICT'S

SUBJECT	Physical Science: Paper 1	DATE	13 September 2021
GRADE	11	MARKS	165 Marks
EXAMINER	Mr F. McCoy	MODERATOR	Mrs S. Govender
NAME		DURATION	2 ½ hours
CLASS			

LOW ORDER 50%

HIGH ORDER 50%

INSTRUCTIONS

1. This question paper consists of 12 pages, an Answer Sheet of 2 pages and a Data Sheet of 2 pages. Please make sure that your question paper is complete.
2. Answer ALL the questions.
3. Read the questions carefully.
4. Use the data and formulae sheet whenever necessary.
5. Start each question on a new page.
6. **Show your working in all calculations.**
7. Units need not be included in the working of calculations, but appropriate units should be shown in all answers.
8. Express all answers to TWO decimal places unless told otherwise.
9. It is in your own interest to write legibly and to present your work neatly.

Question	1	2	3	4	5	6	7	TOTAL
Total	16	26	25	27	29	16	26	165

QUESTION 1

16 MARKS

Answer the following multiple-choice questions on your separate ANSWER SHEET. Make a cross (X) in the box corresponding to the letter which you consider to be the most correct.

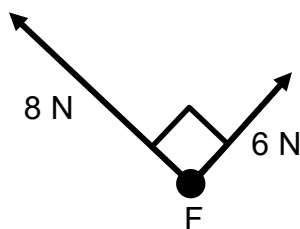
1.1 In the equation of motion $v^2 = u^2 + 2as$, the unit of the term “ $2as$ ” is:

- A $m.s^{-1}$
 - B $m.s^{-2}$
 - C $m^2.s^{-1}$
 - D $m^2.s^{-2}$
- (2)

1.2 Two vectors act simultaneously on the same point. What should the angle between them be so that a maximum resultant is obtained?

- A 0°
 - B 45°
 - C 90°
 - D 180°
- (2)

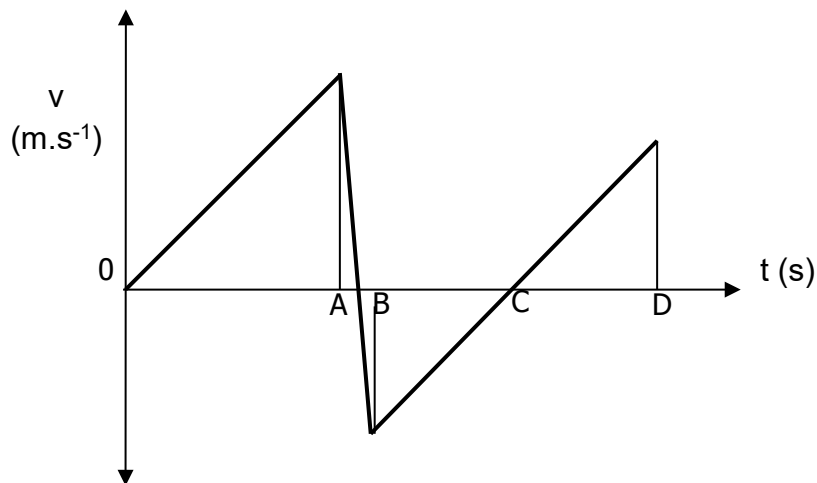
1.3 The vector diagram shows two forces in the same plane acting on an object F.



Another 10 N force, in the same plane as the other forces, is applied on the object F. Which of the following represents the direction at which the 10 N force must be applied to ensure object F is in equilibrium? (2)

A	B
C	D

- 1.4 The velocity-time graph below represents a ball which is dropped and bounces once. Which letter (A, B, C or D) gives the time when the ball is at the top of its bounce?



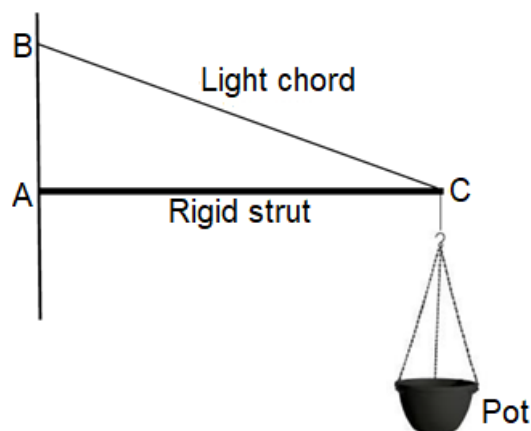
(2)

- 1.5 A constant net force of 1 N acts on a stationary 5 000 kg car. The car will...

- A stay at rest because its friction is greater than 1 N.
- B stay at rest because 1 N is too little to overcome its inertia.
- C move at a constant velocity.
- D accelerate uniformly.

(2)

- 1.6 The diagram below shows a pot of weight 40 N suspended from point C by a rigid strut AC fixed to the wall at A and a light chord BC attached to the wall at B.



Which statement about the forces acting on point C is true?

- A The vertical component of the tension in chord BC is equal to 40 N upwards.
- B The tension in chord BC must be 40 N.
- C The tension in cord BC is less than 40 N.
- D The vector sum of the three forces acting on C must be 40 N.

(2)

- 1.7 A piano moving company have found themselves in some trouble because a grand piano on a platform with wheels managed to roll out of the back of one of their trucks and was badly damaged.

An eye witness said that the back doors of the truck were not properly closed but the driver moved the truck forward and the piano (on its platform) came rolling out the back of the truck.

Which one of the following options correctly explains why the piano rolled out the back of the truck?

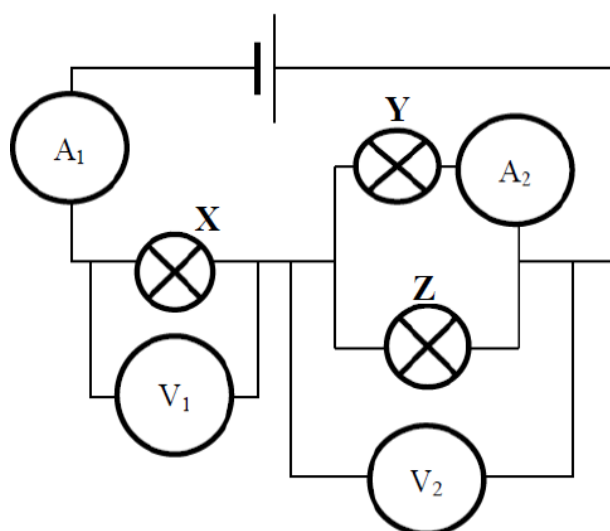
	Explanation	Applicable Law of motion
A	There was insufficient force applied to the piano and no net force resulted, so the piano did not accelerate forward.	Newton's First law
B	The force of the piano on the truck caused an equal but opposite force on the piano and pushed it out the truck.	Newton's First law
C	There was insufficient force applied to the piano and no net force resulted, so the piano did not accelerate forward.	Newton's Third law
D	The force of the piano on the truck caused an equal but opposite force on the piano and pushed it out the truck.	Newton's Third law

(2)

- 1.8 In the circuit shown in the diagram alongside the bulbs are all identical. The resistances of the ammeters, connecting wires and cell are negligible.

Which one of the following correctly gives the ratio of the power dissipated in bulb X (P_X) to the power dissipated in bulb Y (P_Y)?

	$P_X:P_Y$
A	4:1
B	1:4
C	2:1
D	1:2

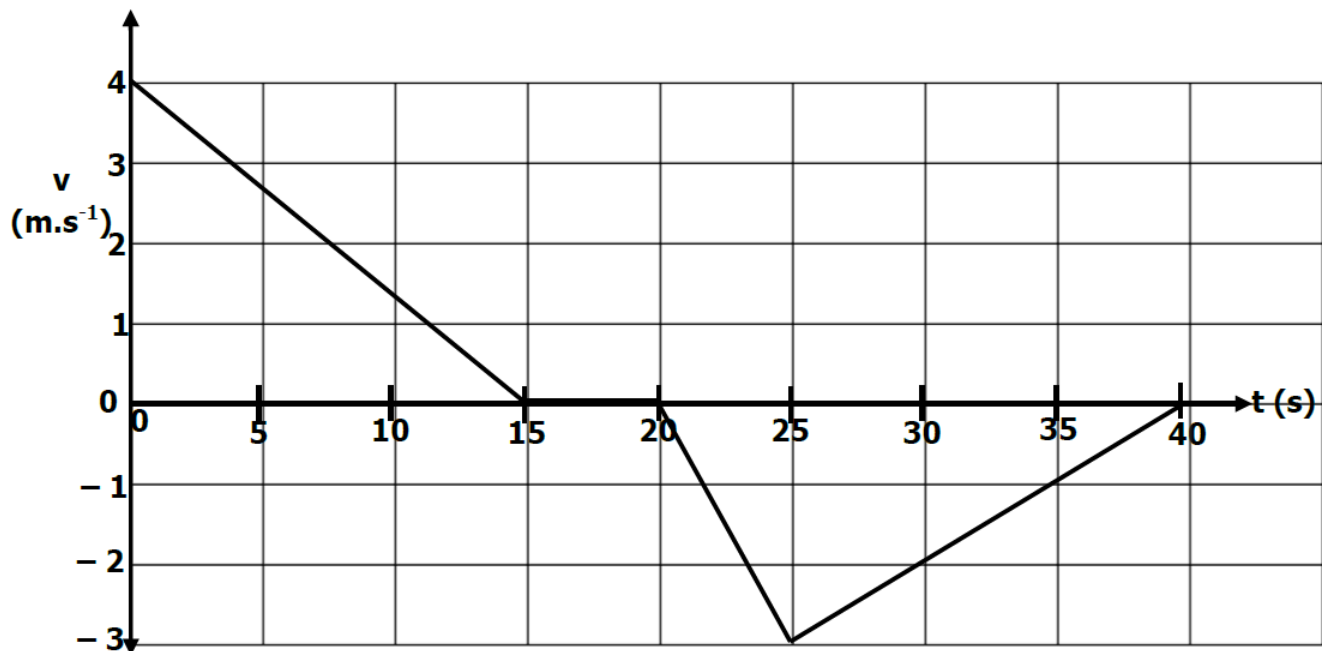


(2)

QUESTION 2

26 MARKS

The variation with time t of the velocity v of a car, initially traveling west is shown in the graph below.



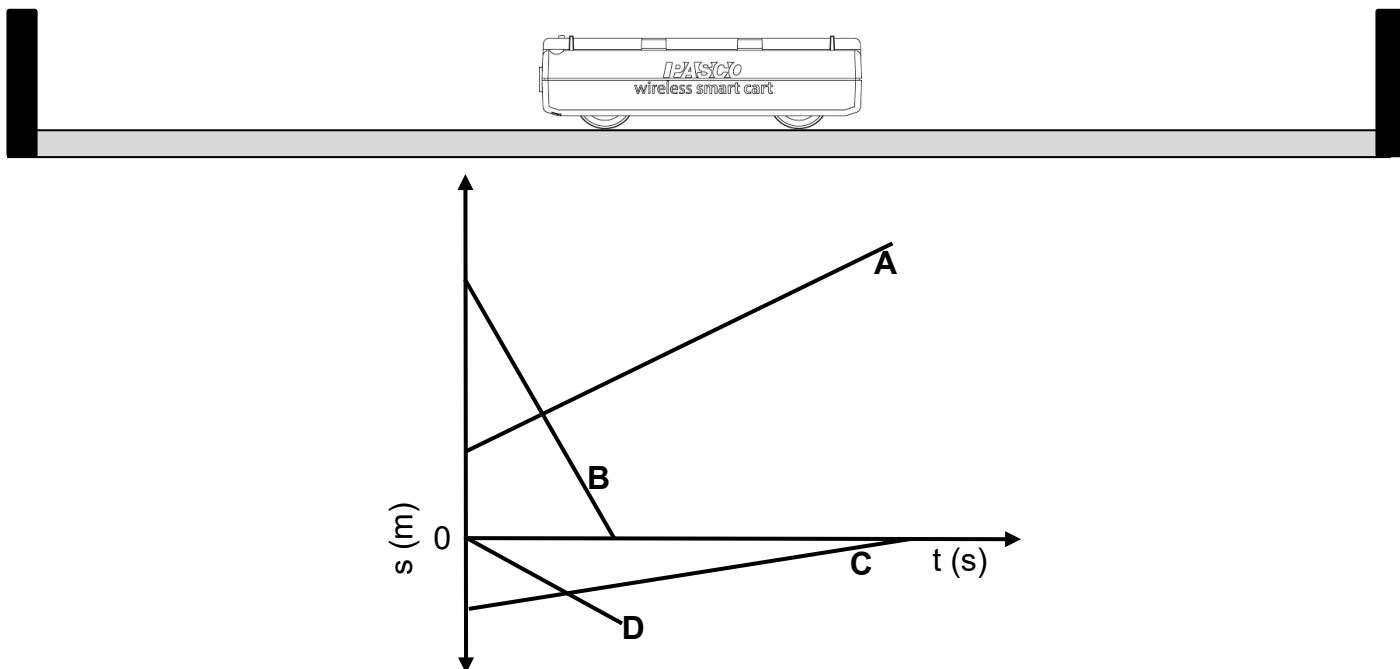
- 2.1 Define *velocity*. (2)
- 2.2 During which time interval(s) is the car's velocity constant? (1)
- 2.3 Describe how the velocity of the car is changing between 20 s and 25 s. (3)
- 2.4 Define *acceleration*. (2)
- 2.5 Determine the acceleration of the car between 25 s and 40 s. (4)
- 2.6 At which time/s is the car the greatest distance from the origin? (2)
- 2.7 Use the graph of motion to determine the average speed of the car between 0 and 40 s. (4)
- 2.8 Will the magnitude of the average velocity of the car between 0 and 40 s be GREATER THAN, LESS THAN or EQUAL TO the magnitude of the average speed of the car during the same time interval? Explain your answer by demonstrating an understanding of the relevant physics concepts. No further calculations are required. (4)
- 2.9 Sketch an acceleration vs. time graph for the car from 0 s to 40 s. Acceleration values need NOT be shown. (4)

QUESTION 3

25 MARKS

3.1 The displacement vs time graph shows the motion a wireless smart cart sliding in a straight line across a horizontal track.

Right is taken as positive and the middle point is taken as the point of reference.



- 3.1.1 Which graph shows the cart starting from the middle of the track? (2)
- 3.1.2 Which graph shows the cart moving with the smallest constant velocity? (2)
- 3.1.3 Which graph shows the cart moving right away from the middle of the track? (2)
- 3.1.4 Which graph shows the cart moving left towards the middle of the track? (2)
- 3.2 A red cart is moving at a constant speed of $0,3 \text{ m.s}^{-1}$ towards the end of a track that is 30 cm away. The cart is 16,5 cm in length.
- 3.2.1 Define *distance*. (1)
- 3.2.2 How long does it take for the red cart to reach the end of the track? (4)
- A blue cart is 50 cm behind the red cart, moving at a constant speed of $0,4 \text{ m.s}^{-1}$ towards the red cart.
- 3.2.3 How far does the blue cart move in the time it takes the red cart to reach the end of the track? (2)
- 3.2.4 How far is the blue cart from the end of the track when the red cart reaches the end of the track? (2)
- 3.2.5 What must the acceleration of the blue cart be, so that it touches the back of the red cart at the moment the red cart reaches the end of the track? (4)
- 3.3 A cart starts from rest and rolls along a track, a distance L with uniform acceleration, reaching speed v . Determine the length of the track needed in terms of L , if you want the cart to have a speed $2v$ at the end of the track. (4)

QUESTION 4

27 MARKS

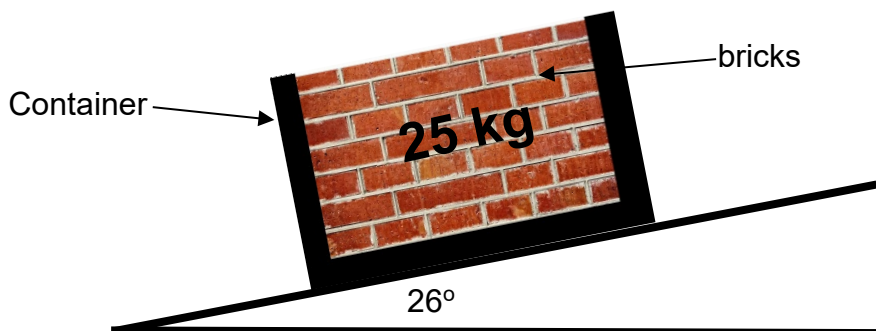
A toy rocket of mass 0,5 kg is placed vertically in a bottle and ignited. It accelerates vertically upwards at 8 m.s^{-2} . It experiences air resistance of 0,5 N for the entire flight.

- 4.1 Define *weight*. (2)
 - 4.2 Name and describe the reaction force to the weight of the rocket. (2)
 - 4.3 Calculate the upwards force required to accelerate the rocket vertically at 8 m.s^{-2} . (4)
 - 4.4 Calculate how far the rocket travels in 3 seconds. (3)
 - 4.5 After 3 seconds of flight a 125 g piece of the nose-cone falls off and as it falls it experiences a 0,1 N frictional force.
 - 4.5.1 Define *friction force due to a surface*. (2)
 - 4.5.2 Calculate the velocity of the nose-cone piece as it falls off. (3)
 - 4.5.3 Draw a free-body diagrams for the nose cone when it is moving:
 - a) up; (2)
 - b) and down. (2)
 - 4.5.4 Calculate the net force acting on this piece when it is moving down. (3)
 - 4.5.5 Sketch v-t graph for the motion of the nose piece from the moment it falls off the rocket until it hits the ground. (4)
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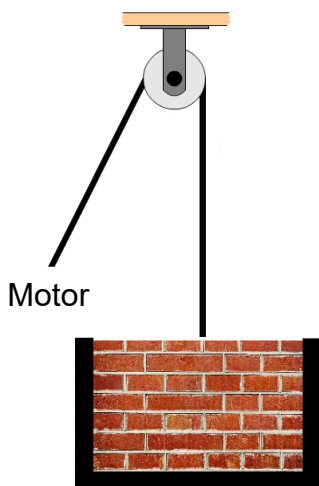
QUESTION 5

29 MARKS

- 5.1 A container of bricks of mass 25 kg is just on the point of slipping when placed on a plane inclined at 26° to the horizontal, as shown in the diagram below.



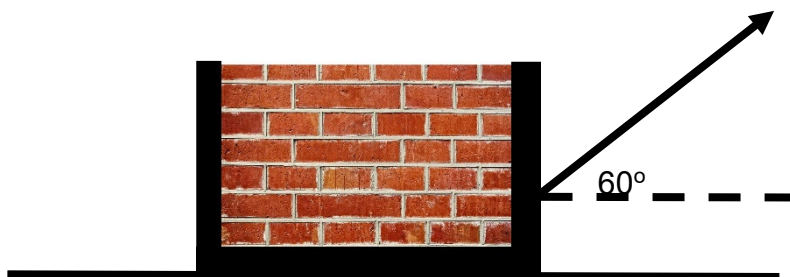
- 5.1.1 State *Newton's first law*. (2)
- 5.1.2 According to Newton's first law what is the magnitude of the net force acting on the container of bricks while it is at rest on the plane? (1)
- 5.1.3 Calculate the magnitude of the maximum static frictional force acting on the container of bricks. (3)
- 5.1.4 Define *normal force*. (2)
- 5.1.5 Calculate the magnitude of the normal force acting on the container of bricks. (3)
- 5.1.6 Calculate the co-efficient of static friction between the container and the surface of the plane. (2)
- 5.2 A motor is used to move the container of bricks, of mass 25 kg, vertically upwards, as shown in the diagram below.



The container of bricks is accelerated upwards at $0,4 \text{ m.s}^{-2}$. The effects of air resistance can be ignored.

- 5.2.1 Calculate the magnitude of the tension in the rope lifting the container of bricks. (4)
- 5.2.2 What would the normal force be on the bricks, if the rope snapped? Explain your answer. (2)

- 5.3 The container of bricks (of mass 25 kg) is now placed on a rough horizontal surface and is pulled by a cable which is inclined at 60° to the horizontal as shown in the diagram. The co-efficient of static friction between the container and the surface is 0,72.

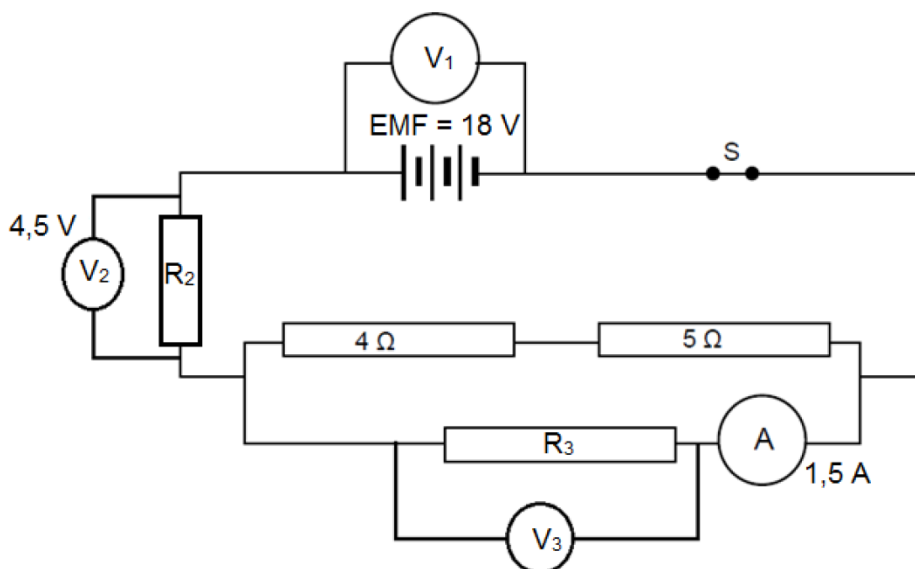


- 5.3.1 Draw a labelled free-body diagram showing all of the forces acting on the container. (4)
- 5.3.2 Calculate the minimum force that the cable must exert on the container in order to just overcome the force of maximum static friction. (6)
-

QUESTION 6

16 MARKS

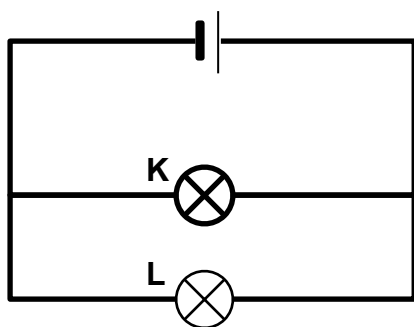
- 6.1 Consider the circuit below, in which the cells and wires have no internal resistance, and the meters have no unwanted effect on the circuit.



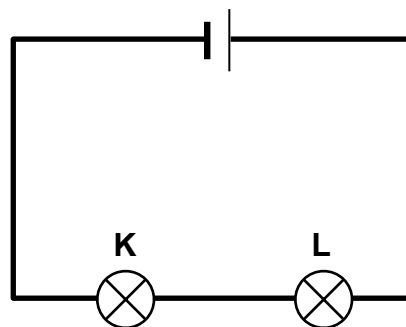
When the switch is closed V_1 reads 18 V, V_2 reads 4,5 V, and the ammeter reads 1,5 A.

- 6.1.1 Calculate how much charge passes through R_3 in 1 minute. (3)
- 6.1.2 Write down the reading on V_3 . (2)
- 6.1.3 Calculate the resistance of R_3 . (3)
- 6.1.4 The resistor R_3 is removed. What will happen to the reading on V_2 ? Explain your answer, making reference to a relevant equation. (4)
- 6.2 The same cell with no internal resistance is connected in turn to each of the two circuits shown below. Bulb **K** has twice the resistance of Bulb **L**.

Circuit 1



Circuit 2



How does the brightness of Bulb **K** compare with that of Bulb **L** in each of these circuits? Explain your answer (4)

QUESTION 7

26 MARKS

A simulation is used to prove Newton's law of gravitation. Represented by the equations below:

$$F = \frac{Gm_1m_2}{r^2}$$

Where:

F = the Gravitational force between the objects (N)

m_1, m_2 = the masses of the two objects (kg)

r = the distance between the object's centres (m)

G = Constant of Universal Gravitation. ($6,7 \times 10^{-11} \text{N} \cdot \text{m}^2 \cdot \text{kg}^{-2}$)

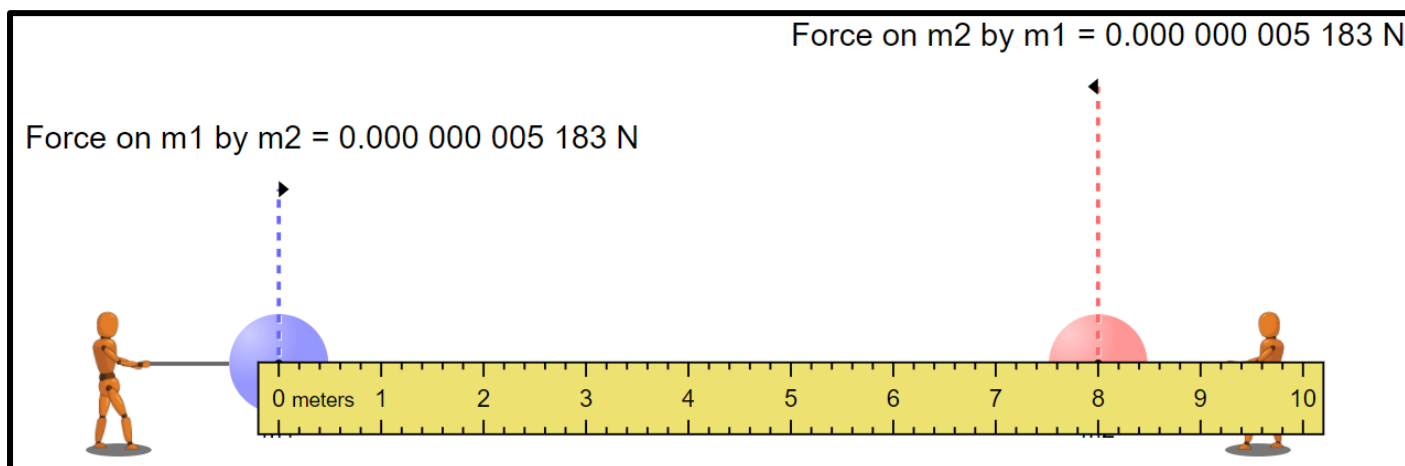
The table below shows the gravitational force of attraction between 2 identical spheres when placed different distances apart.

Distance between centres (m)	Force experienced ($\times 10^{-8} \text{ N}$)	Inverse of distance between centres squared (m^{-2})
4	2,073	0,063
5	1,327	0,040
6	0,921	0,027
7	0,677	0,020
8		

7.1 The diagram below shows the masses at 8 meters apart.

7.1.1 Read off the force experienced and write it in the table on the ANSWER SHEET. (2)

7.1.2 Calculate the value in the last column write it in the table on the ANSWER SHEET. (2)



- 7.2 Plot a graph of inverse of distance squared vs force on the ANSWER SHEET.
YOU MUST USE THE SCALE THAT HAS BEEN GIVEN. (7)
- 7.3 Write the relationship between these variables in words. (2)
- 7.4 Use your graph to determine the force these spheres would experience if they were
4,5 m apart from each other. Show how you determined this on your graph. (3)
- 7.5 Calculate the gradient of the graph. Include a unit. (4)
- 7.6 What physical quantities are represented by the gradient? (3)
- 7.7 Hence, use your answer in 7.5 and 7.6 to determine the mass of each spheres. (3)

END OF EXAMINATION



PHYSICAL SCIENCE PAPER 1

GRADE 11

13 SEPTEMBER 2021

ST BENEDICT'S

ANSWER SHEET

NAME: _____

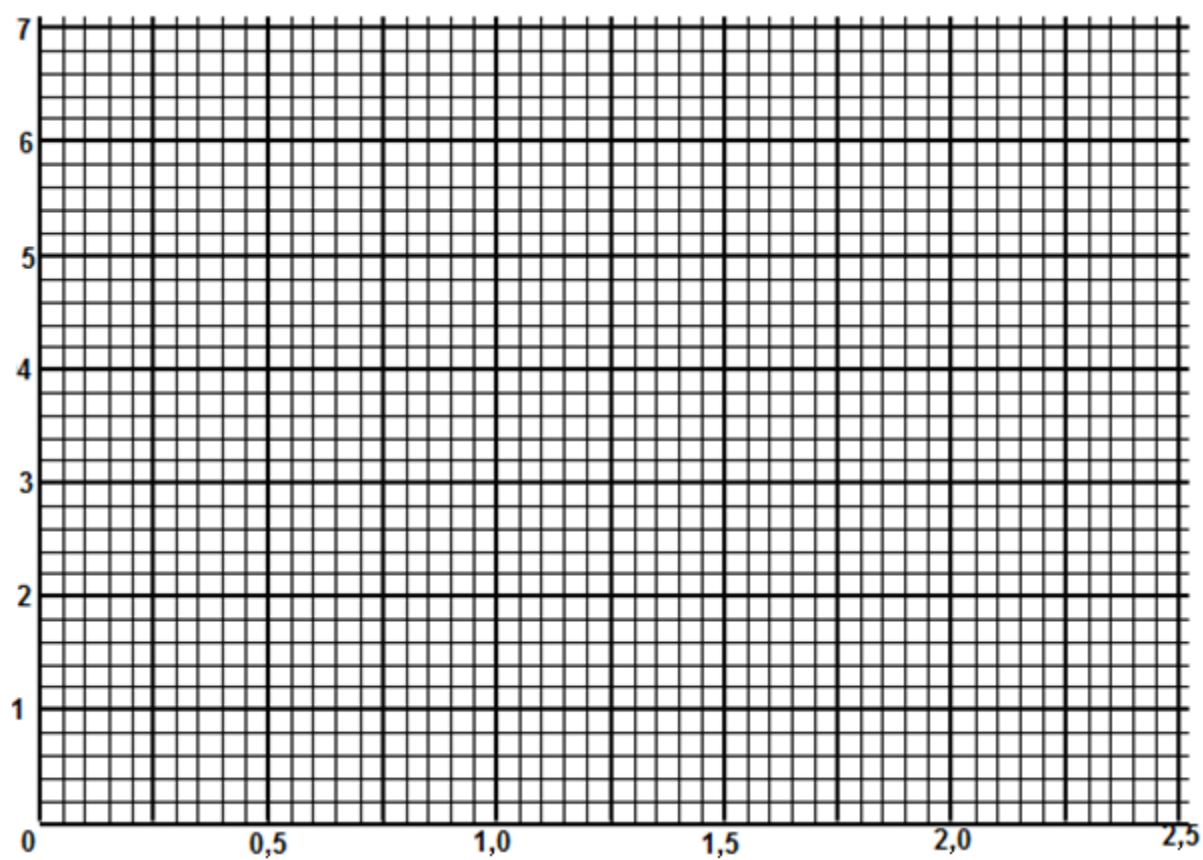
QUESTION 1

1.1	A	B	C	D
1.2	A	B	C	D
1.3	A	B	C	D
1.4	A	B	C	D
1.5	A	B	C	D
1.6	A	B	C	D
1.7	A	B	C	D
1.8	A	B	C	D

QUESTION 7.1

Distance between centres (m)	Force experienced ($\times 10^{-8}$ N)	Inverse of distance between centres squared (m^{-2})
4	2,073	0,063
5	1,327	0,040
6	0,921	0,027
7	0,677	0,020
8		

QUESTION 7.2



EXAMINATION DATA SHEET FOR THE PHYSICAL SCIENCES (PHYSICS)

TABLE 1 PHYSICAL CONSTANTS

NAME	SYMBOL	VALUE
Acceleration due to gravity	g	$9,8 \text{ m}\cdot\text{s}^{-2}$
Speed of light in a vacuum	c	$3,0 \times 10^8 \text{ m}\cdot\text{s}^{-1}$
Universal gravitational constant	G	$6,7 \times 10^{-11} \text{ N}\cdot\text{m}^2\cdot\text{kg}^{-2}$
Coulomb's constant	k	$9,0 \times 10^9 \text{ N}\cdot\text{m}^2\cdot\text{C}^{-2}$
Magnitude of charge on electron	e	$1,6 \times 10^{-19} \text{ C}$
Mass of an electron	m_e	$9,1 \times 10^{-31} \text{ kg}$
Planck's constant	h	$6,6 \times 10^{-34} \text{ J}\cdot\text{s}$
1 electron volt	eV	$1,6 \times 10^{-19} \text{ J}$

TABLE 2 PHYSICS FORMULAE

MOTION

$v = u + at$ or $v_f = v_i + a\Delta t$	$s = \left(\frac{v+u}{2}\right)t$ or $\Delta x = \left(\frac{v_f+v_i}{2}\right)\Delta t$
$v^2 = u^2 + 2as$ or $v_f^2 = v_i^2 + 2a\Delta x$	$s = ut + \frac{1}{2}at^2$ or $\Delta x = v_i\Delta t + \frac{1}{2}a(\Delta t)^2$

FORCE AND MOMENTUM

$F_{net} = ma$	$F_{net} = \frac{\Delta p}{\Delta t}$ or $F_{net}\Delta t = m\Delta v$	$J = \Delta p = mv - mu$ or $J = \Delta p = mv_f - mv_i$
$p = mv$	$F_g = mg$	$F_{fs}^{max} = \mu_s F_N$ $F_{fk} = \mu_k F_N$

WORK, ENERGY AND POWER

$W = Fs$ or $W = F\Delta x$ or $W = F\Delta x \cos\theta$	$P = \frac{W}{t}$	$P = Fv$
$E_p = mgh$	$E_k = \frac{1}{2}mv^2$	$W_{net} = \Delta E_K$
efficiency = $\frac{\text{power}_{out}}{\text{power}_{in}} \times 100$		

GRAVITATIONAL AND ELECTRIC FIELDS

$F = G \frac{m_1 m_2}{r^2}$	$g = \frac{F}{m}$	$g = G \frac{M}{r^2}$
$F = k \frac{q_1 q_2}{r^2}$	$E = \frac{F}{q}$	$E = \frac{kQ}{r^2}$

ELECTRIC CIRCUITS

$I = \frac{q}{t}$	$V = \frac{W}{q}$
$R = \frac{V}{I}$	$emf = I(R_{ext} + r)$ or $emf = V_{load} + V_{internal\ resistance}$
$R_s = R_1 + R_2 + \dots$	$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$
$P = \frac{W}{t} \quad \text{or} \quad W = Pt$	
$W = VIt \quad \text{or} \quad W = I^2 R t \quad \text{or} \quad W = \frac{V^2}{R} t$	
$P = VI \quad \text{or} \quad P = I^2 R \quad \text{or} \quad P = \frac{V^2}{R}$	

ELECTRODYNAMICS

$\Phi = BA \cos \theta$	$emf = -\frac{N \Delta \Phi}{\Delta t}$	$F = IB \ell \sin \theta$
$V_p I_p = V_s I_s$	$\frac{N_s}{N_p} = \frac{V_s}{V_p}$	

PHOTONS AND ELECTRONS

$c = f \lambda$		$E = hf$		or	$E = \frac{hc}{\lambda}$
$E = W_0 + E_{K(max)}$	$W_0 = hf_0$		$E_{K(max)} = \frac{1}{2}mv_{max}^2$		